

Task 5 — Insights & Summary

Short paragraph

This research project aimed at predicting house prices using various Machine Learning Algorithms. After going through the data, it can be concluded that there were several variables, which had an influence on house prices, like area, bathrooms, bedrooms, stories, parking, and whether the house is furnished or not. We trained two different models, namely the Linear Regression Model and the Random Forest Regression Model, but the best results were provided by the Linear Regression Model with an R^2 Score of 0.6529, which means that 65% variation in house prices could be explained through Linear Regression. An interesting point here is that the Random Forest model did not perform better than Linear Regression on this particular dataset. It can be suggested that real estate firms should consider important factors for houses while determining their price.

Which features influence house price the most?

The variables which had an impact on the prices of the houses included area, bathrooms, number of storeys, parking, and air conditioning. Houses with better facilities and amenities generally tended to have higher prices.

How accurate was your model?

The Linear Regression model had an R^2 Score of 0.6529. This implies that it managed to explain about 65% of the variation in house prices. Simply put, this model was relatively accurate in its predictions regarding house prices based on available property attributes. The predictions were not perfect but still managed to capture all important trends from the available data set.

What surprised you in the data?

One interesting thing was that the Random Forest Regressor could not outperform the Linear Regression Model in terms of accuracy. The reason for this is that random forest is generally expected to be more accurate since it is able to account for complexities in the data. In this case, however, the Linear Regression Model had a higher R^2 Score and less prediction errors.

One recommendation for a real estate business

From the above analysis, real estate

businesses are advised to pay attention to the size of the plot, the number of bathrooms, the presence of parking space, and other amenities like air conditioners while setting house prices. This is because these variables have shown significant impacts on house prices and hence can be used to create better valuation estimates.